

Fairness Metrics Toolkit

A User Guide

Pedro Salas-Rojo

1 What this is

This document is the user guide for the **Fairness Metrics Toolkit**, a self-contained R implementation of 13 binary (two-group) fairness metrics and 4 multidimensional (N-subgroup) fairness metrics used to audit whether an algorithm's predictions treat a protected group fairly relative to a reference group.

The toolkit is deliberately **metrics only**: it measures fairness, it does not implement or discuss methods to correct an unfair algorithm. It requires no packages beyond base R (**stats**), and ships with a synthetic example dataset so every function can be tried immediately.

1.1 Notation used throughout

- $G \in \{0, 1\}$: protected attribute. $G = 0$ is the disadvantaged group, $G = 1$ the advantaged/reference group.
- $Y \in \{0, 1\}$: true outcome.
- $\hat{Y} \in \{0, 1\}$: the algorithm's decision (denoted **Yhat** in the code).
- $\text{TPR} = P(\hat{Y} = 1 \mid Y = 1)$, $\text{FPR} = P(\hat{Y} = 1 \mid Y = 0)$, $\text{TNR} = P(\hat{Y} = 0 \mid Y = 0)$, $\text{FNR} = P(\hat{Y} = 0 \mid Y = 1)$.
- $\text{PPV} = P(Y = 1 \mid \hat{Y} = 1)$ (precision), $\text{NPV} = P(Y = 0 \mid \hat{Y} = 0)$.
- $\text{Acc} = P(\hat{Y} = Y)$.
- zg_m : subgroup m in an integer-coded partition **ZG** used by the 4 multidimensional metrics (any number of subgroups).
- ε (**eps**): discrimination-aversion parameter for the multidimensional metrics. Larger ε relaxes the fairness threshold; $\varepsilon = 0$ is the strictest case.

2 Repository structure and quick start

```
fairness_metrics/  
|-- README.md  
|-- fairness_functions.R          <- every metric function  
|-- 01_load_data.R              <- loads + validates the data  
|-- 02_compute_metrics_one_by_one.R <- each metric, explained individually  
|-- 03_compute_all_metrics.R     <- every metric at once + summary CSV  
|-- data/mock_data.csv          <- ships pre-built  
'-- docs/fairness_metrics_guide.pdf <- this document
```

`data/mock_data.csv` ships pre-built, so no generation step is needed. From the repository root, run the scripts in order:

```
Rscript 02_compute_metrics_one_by_one.R
Rscript 03_compute_all_metrics.R
```

Running `03_compute_all_metrics.R` writes a summary to `output/fairness_metrics_summary.csv`, created automatically on first run.

To apply the toolkit to your own data, build a data frame with the columns described in Table 1 and call `compute_binary_metrics()` and/or `compute_all_multidimensional()` directly – nothing else in the repository needs to change.

Column	Used by	Requirement
G	13 binary metrics	exactly 2 values (any coding)
Y	13 binary metrics	true outcome, binary (0/1)
Yhat	13 binary metrics	algorithm’s decision, binary (0/1)
A	Conditional Equal Opportunity (optional)	any categorical variable
ZG	4 multidimensional metrics	integer subgroup code 0, 1, 2, ...

Table 1: Data requirements to reuse the toolkit on new data.

3 The worked example: a loan-approval algorithm

`data/mock_data.csv` was built by simulating 5,000 loan applicants. Each has covariates (income, credit score, employment years) that determine whether they would truly repay a loan (Y), and an algorithm predicts a repayment probability (`score`) and approves the loan if that probability clears 0.5 ($\hat{Y}$). The simulation builds in two sources of disparity across the protected attribute G ($0 =$ disadvantaged, $1 =$ advantaged):

1. a structural gap in the applicant covariates themselves, reflecting a real-world disparity upstream of any algorithm; and
2. a direct algorithmic penalty against $G = 0$ on top of that – the kind of bias a fairness audit is meant to catch.

A second attribute, `region`, is crossed with G into four subgroups ($ZG = 2 \cdot G + \text{region} \in \{0, 1, 2, 3\}$) used by the multidimensional metrics, and `age_band` is used to illustrate Conditional Equal Opportunity. No real data is used anywhere in this repository. With seed 123, the resulting sample has:

	$G = 0$ (disadvantaged)	$G = 1$ (advantaged)
N	2,255	2,745
True repayment rate, $P(Y = 1)$	0.356	0.564
Approval rate, $P(\hat{Y} = 1)$	0.251	0.616

All numbers reported in the rest of this guide are computed directly from this dataset by `03_compute_all_metrics.R`.

4 The 13 binary fairness metrics

Metrics ending in `_diff` have expected value 0 under parity; metrics ending in `_ratio` have expected value 1. Every metric compares $G = 0$ against $G = 1$.

Metric	Formula	Value	Reference
Statistical Parity (<code>SP_diff</code>)	$P(\hat{Y}=1 G=0) - P(\hat{Y}=1 G=1)$	-0.365	Dwork et al. (2012)
Disparate Impact (<code>DI_ratio</code>)	$P(\hat{Y}=1 G=0)/P(\hat{Y}=1 G=1)$	0.408	Feldman et al. (2015)
Equal Opp., (<code>EqOp_plus_diff</code>)	TPR TPR(0) - TPR(1)	-0.296	Hardt et al. (2016)
Equal Opp., (<code>EqOp_minus_diff</code>)	TNR TNR(0) - TNR(1)	0.298	Hardt et al. (2016)
Eq. Odds, (<code>EqOdds_plus_diff</code>)	TPR TPR(0) - TPR(1)	-0.296	Hardt et al. (2016)
Eq. Odds, (<code>EqOdds_minus_diff</code>)	FPR FPR(0) - FPR(1)	-0.298	Hardt et al. (2016)
Treatment Equality, diff	$(\text{FPR}/\text{FNR})(0) - (\text{FPR}/\text{FNR})(1)$	-1.536	Berk et al. (2021)
Treatment Equality, ratio	$(\text{FPR}/\text{FNR})(0)/(\text{FPR}/\text{FNR})(1)$	0.140	Berk et al. (2021)
Overall Acc., diff	$\text{Acc}(0) - \text{Acc}(1)$	0.047	Berk et al. (2021)
Overall Acc., ratio	$\text{Acc}(0)/\text{Acc}(1)$	1.070	Berk et al. (2021)
Equalized Disincentive	$[\text{TPR}-\text{FPR}](0) - [\text{TPR}-\text{FPR}](1)$	0.003	Youden (1950)
PPV parity	$\text{PPV}(0) - \text{PPV}(1)$	-0.039	Chouldechova (2017)
NPV parity	$\text{NPV}(0) - \text{NPV}(1)$	0.100	Chouldechova (2017)

Table 2: The 13 binary fairness metrics, with the value each takes on the mock loan-approval example (eps not applicable). Full citations in Section 6.

4.1 How to read the worked example

The algorithm approves loans for group $G = 0$ far less often than for $G = 1$ (Statistical Parity = -0.365; Disparate Impact = 0.408, well below the conventional 0.8 “four-fifths rule” threshold). Part of this gap reflects a real difference in creditworthiness across groups (the true repayment rate is 0.356 vs. 0.564), but Equal Opportunity (-0.296) shows that even among applicants who *would* repay, $G = 0$ is approved substantially less often – evidence of disparate treatment beyond what the underlying risk alone would justify. Equalized Odds fails on both components (TPR and FPR), so Equal Opportunity alone is not being satisfied “by accident”.

Interestingly, Overall Accuracy is nearly identical across groups (in fact slightly *higher* for $G = 0$, diff = 0.047) and Equalized Disincentive (the group-level Youden’s J gap) is essentially zero (0.003). This is the guide’s central methodological point: **a single metric is not enough**. Accuracy parity, or a near-zero Youden’s-J gap, can coexist with large gaps in Statistical Parity, Equal Opportunity, Disparate Impact, PPV and NPV – each metric encodes a different normative notion of what “fair” means, and they are not implied by one another (Kleinberg, Mullainathan & Raghavan, 2016, formalize exactly this kind of incompatibility).

Finally, PPV parity (-0.039) and NPV parity (0.100) show that among *approved* applicants, those from $G = 0$ are somewhat less likely to actually repay than approved $G = 1$ applicants – the approval decision is less “reliable” when applied to $G = 0$.

4.2 Conditional Equal Opportunity

Conditional Equal Opportunity re-computes the Equal Opportunity TPR gap separately within each level of a third variable (here, `age_band`), which is useful to check whether an aggregate gap is driven by one stratum or is pervasive across all of them. This is an extension implemented in this toolkit, building on Hardt et al. (2016); it is not itself a metric with a single canonical source paper.

Age band	TPR($G=0$)	TPR($G=1$)	CEO gap
18–30	0.445	0.758	−0.313
31–50	0.474	0.753	−0.279
51+	0.457	0.765	−0.307

The gap is essentially constant across strata (≈ -0.30 everywhere), so the aggregate Equal Opportunity violation is pervasive, not driven by a single age group.

5 The 4 multidimensional fairness metrics

These extend fairness beyond two groups to an arbitrary number of subgroups $\mathbf{ZG}$. All four are governed by a discrimination-aversion parameter ε ; the values below use $\varepsilon = 0.10$ on the four $\mathbf{ZG}$ subgroups of the mock data ($\mathbf{ZG} = 2 \cdot G + \mathbf{region}$).

Metric		Formula	Value	Fair?	Reference
Statistical Parity (SubPar)	Subgroup	$\max_m [P(zg_m) \cdot P(\hat{Y}=1) - P(\hat{Y}=1 zg_m) - \varepsilon]$	0.049	Yes	Kearns et al. (2018)
False Positive Parity (FalPos)	Subgroup	$\max_m [P(Y=0, zg_m) \cdot \text{FPR}_{\text{all}} - \text{FPR}_{zg_m} - \varepsilon]$	0.021	Yes	Kearns et al. (2018)
Differential Fairness, $c = 1$ (DiffFair)	Fairness,	$\max_m P(\hat{Y}=c zg_m) / \min_m P(\hat{Y}=c zg_m) - e^\varepsilon$	2.606	No	Foulds et al. (2020)
Differential Fairness, $c = 0$	Fairness,	(same formula, $c = 0$)	2.044	No	Foulds et al. (2020)
Worst-Case Fairness, $c = 1$ (WorCas)	Fairness,	$1 - \min_m P(\hat{Y}=c zg_m) / \max_m P(\hat{Y}=c zg_m)$	0.616	No	Ghosh et al. (2021)
Worst-Case Fairness, $c = 0$	Fairness,	(same formula, $c = 0$)	0.511	No	Ghosh et al. (2021)

Table 3: The 4 multidimensional fairness metrics on the mock data ($\varepsilon = 0.10$). “Fair?” reports whether the metric’s own fairness condition holds at that ε .

Note the disagreement between the two “subgroup” metrics (SubPar, FalPos – both pass at $\varepsilon = 0.10$) and the two “ratio”-based metrics (DiffFair, WorCas – both fail). SubPar and FalPos weight each subgroup’s deviation by its population share, so a disparity concentrated in a smaller subgroup contributes less to the global statistic; DiffFair and WorCas instead look at the most extreme ratio across *all* subgroups regardless of size, and are far more sensitive to exactly this kind of concentrated disparity (the two largest $\mathbf{ZG}$ subgroups here, 2 and 3, have approval rates around 0.60–0.63 vs. around 0.24–0.26 for subgroups 0 and 1). This is the same lesson as Section 3.1 at a different level: which metric you use determines what kind of unfairness you can detect.

6 References

- Berk, R., Heidari, H., Jabbari, S., Kearns, M., & Roth, A. (2021). Fairness in Criminal Justice Risk Assessments: The State of the Art. *Sociological Methods & Research*, 50(1), 3–44.
- Chouldechova, A. (2017). Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments. *Big Data*, 5(2), 153–163.
- Dwork, C., Hardt, M., Pitassi, T., Reingold, O., & Zemel, R. (2012). Fairness Through Awareness. *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference (ITCS)*.
- Feldman, M., Friedler, S. A., Moeller, J., Scheidegger, C., & Venkatasubramanian, S. (2015). Certifying and Removing Disparate Impact. *Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 259–268.
- Foulds, J. R., Islam, R., Keya, K. N., & Pan, S. (2020). An Intersectional Definition of Fairness. *2020 IEEE 36th International Conference on Data Engineering (ICDE)*.
- Ghosh, A., Genuit, L., & Reagan, M. (2021). Characterizing Intersectional Group Fairness with Worst-Case Comparisons. *Proceedings of the 2nd Workshop on Diversity in Artificial Intelligence (DAI)*.
- Hardt, M., Price, E., & Srebro, N. (2016). Equality of Opportunity in Supervised Learning. *Advances in Neural Information Processing Systems (NeurIPS)*.
- Kearns, M., Neel, S., Roth, A., & Wu, Z. S. (2018). Preventing Fairness Gerrymandering: Auditing and Learning for Subgroup Fairness. *Proceedings of the 35th International Conference on Machine Learning (ICML)*. Companion empirical study: Kearns, Neel, Roth & Wu, An Empirical Study of Rich Subgroup Fairness for Machine Learning, *FAT* 2019*.
- Kleinberg, J., Mullainathan, S., & Raghavan, M. (2016). Inherent Trade-Offs in the Fair Determination of Risk Scores. *arXiv:1609.05807*.
- Youden, W. J. (1950). Index for Rating Diagnostic Tests. *Cancer*, 3(1), 32–35.

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How to cite this toolkit

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